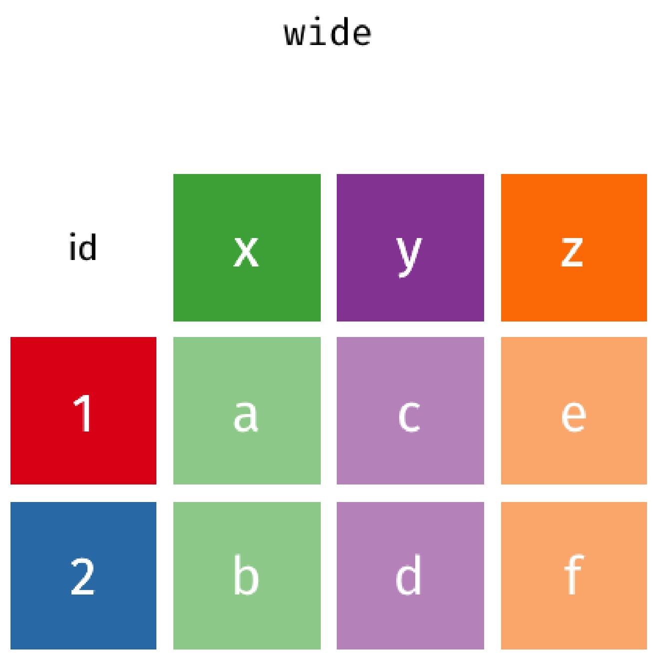
Data Reshaping

Data Wrangling in R

Reshaping: wide vs. long data

https://github.com/gadenbuie/tidyexplain/blob/main/images/tidyr-pivoting.gif



What is wide/long data?

Data is stored *differently* in the tibble.

Wide: has many columns

```
# A tibble: 1 × 4
  State   June_vacc_rate May_vacc_rate April_vacc_rate
  <chr>            <dbl>         <dbl>           <dbl>
1 Alabama          0.516         0.514           0.511
```

Long: column names become data

```
# A tibble: 3 × 3
  State   name            value
  <chr>   <chr>           <dbl>
1 Alabama June_vacc_rate  0.516
2 Alabama May_vacc_rate   0.514
3 Alabama April_vacc_rate 0.511
```

What is wide/long data?

Wide: multiple columns per individual, values spread across multiple columns

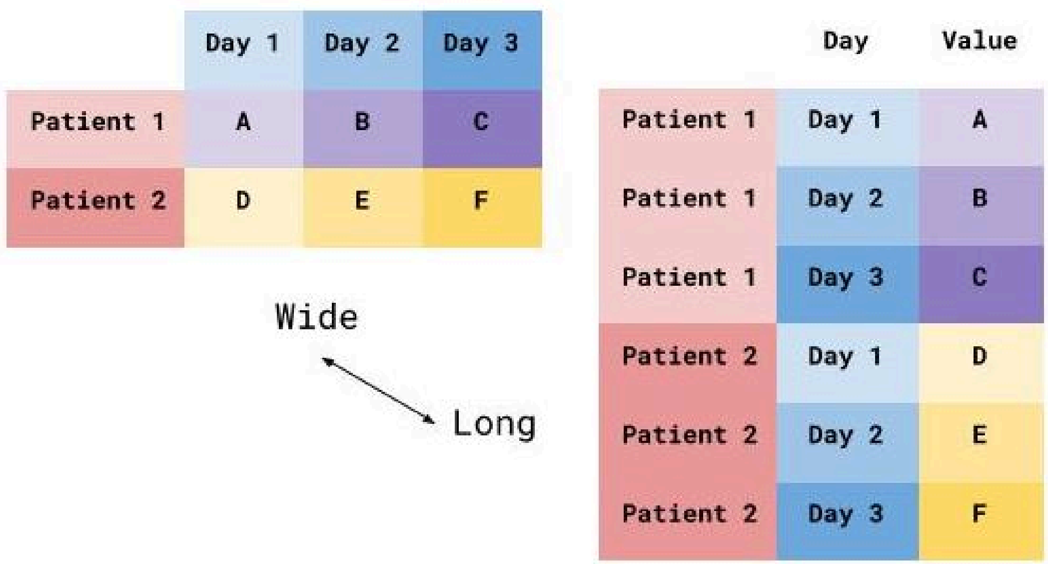
```
# A tibble: 2 × 4
  State   June_vacc_rate May_vacc_rate April_vacc_rate
  <chr>            <dbl>         <dbl>           <dbl>
1 Alabama          0.516         0.514           0.511
2 Alaska           0.627         0.626           0.623
```

Long: multiple rows per observation, a single column contains the values

```
# A tibble: 6 × 3
  State   name            value
  <chr>   <chr>           <dbl>
1 Alabama June_vacc_rate  0.516
2 Alabama May_vacc_rate   0.514
3 Alabama April_vacc_rate 0.511
4 Alaska  June_vacc_rate  0.627
5 Alaska  May_vacc_rate   0.626
6 Alaska  April_vacc_rate 0.623
```

What is wide/long data?

Data is wide or long **with respect** to certain variables.



Why do we need to switch between wide/long data?

Wide: **Easier for humans to read**

```
# A tibble: 2 × 4
  State   June_vacc_rate May_vacc_rate April_vacc_rate
  <chr>            <dbl>         <dbl>           <dbl>
1 Alabama          0.516         0.514           0.511
2 Alaska           0.627         0.626           0.623
```

Long: **Easier for R to make plots & do analysis**

```
# A tibble: 6 × 3
  State   name            value
  <chr>   <chr>           <dbl>
1 Alabama June_vacc_rate  0.516
2 Alabama May_vacc_rate   0.514
3 Alabama April_vacc_rate 0.511
4 Alaska  June_vacc_rate  0.627
5 Alaska  May_vacc_rate   0.626
6 Alaska  April_vacc_rate 0.623
```

Pivoting using `tidyr` package

`tidyr` allows you to “tidy” your data. We will be talking about:

- `pivot_longer` - make multiple columns into variables, (wide to long)
- `pivot_wider` - make a variable into multiple columns, (long to wide)
- `separate` - string into multiple columns

The `reshape` command exists. Its arguments are considered more confusing, so we don’t recommend it.

You might see old functions `gather` and `spread` when googling. These are older iterations of `pivot_longer` and `pivot_wider`, respectively.

pivot_longer...

Reshaping data from **wide to long**

`pivot_longer()` - puts column data into rows (`tidyr` package)

- First describe which columns we want to "pivot_longer"

```
{long_data} <- {wide_data} %>% pivot_longer(cols = {columns to pivot})
```

Reshaping data from **wide to long**

```
wide_data
```

```
# A tibble: 1 × 3
  June_vacc_rate May_vacc_rate April_vacc_rate
           <dbl>         <dbl>           <dbl>
1          0.516         0.514           0.511
```

```
long_data <- wide_data %>% pivot_longer(cols = everything())
long_data
```

```
# A tibble: 3 × 2
  name           value
  <chr>          <dbl>
1 June_vacc_rate  0.516
2 May_vacc_rate   0.514
3 April_vacc_rate 0.511
```

Reshaping data from **wide to long**

`pivot_longer()` - puts column data into rows (`tidyr` package)

- First describe which columns we want to “pivot_longer”
- `names_to =` gives a new name to the pivoted columns
- `values_to =` gives a new name to the values that used to be in those columns

```
{long_data} <- {wide_data} %>% pivot_longer(cols = {columns to pivot},
                                        names_to = {New column name: contains old column names},
                                        values_to = {New column name: contains cell values})
```

Reshaping data from **wide to long**

```
wide_data
```

```
# A tibble: 1 × 3
  June_vacc_rate May_vacc_rate April_vacc_rate
           <dbl>         <dbl>           <dbl>
1          0.516         0.514           0.511
```

```
long_data <- wide_data %>% pivot_longer(cols = everything(),
                                        names_to = "Month",
                                        values_to = "Rate")
long_data
```

```
# A tibble: 3 × 2
  Month            Rate
  <chr>           <dbl>
1 June_vacc_rate  0.516
2 May_vacc_rate   0.514
3 April_vacc_rate 0.511
```

Newly created column names are enclosed in quotation marks.

Data used: Charm City Circulator

https://sisbid.github.io/Data-Wrangling/data/Charm_City_Circulator_Ridership.csv

```
circ <-
  read_csv("https://sisbid.github.io/Data-Wrangling/data/Charm_City_Circulator_Ridership.csv")
head(circ, 5)
```

```
# A tibble: 5 × 15
  day       date  orangeBoardings orangeAlightings orangeAverage purpleBoardings
  <chr>     <chr>           <dbl>            <dbl>         <dbl>           <dbl>
1 Monday    01/1…             877             1027          952               NA
2 Tuesday   01/1…             777              815          796               NA
3 Wednesday 01/1…            1203             1220         1212.              NA
4 Thursday  01/1…            1194             1233         1214.              NA
5 Friday    01/1…            1645             1643         1644               NA
# ℹ 9 more variables: purpleAlightings <dbl>, purpleAverage <dbl>,
#   greenBoardings <dbl>, greenAlightings <dbl>, greenAverage <dbl>,
#   bannerBoardings <dbl>, bannerAlightings <dbl>, bannerAverage <dbl>,
#   daily <dbl>
```

Reshaping data from **wide to long**

```
long <- circ %>%
  pivot_longer(starts_with(c("orange","purple","green","banner")))
long
```

```
# A tibble: 13,752 × 5
   day    date       daily name              value
   <chr>  <chr>      <dbl> <chr>             <dbl>
 1 Monday 01/11/2010   952 orangeBoardings     877
 2 Monday 01/11/2010   952 orangeAlightings   1027
 3 Monday 01/11/2010   952 orangeAverage       952
 4 Monday 01/11/2010   952 purpleBoardings      NA
 5 Monday 01/11/2010   952 purpleAlightings     NA
 6 Monday 01/11/2010   952 purpleAverage        NA
 7 Monday 01/11/2010   952 greenBoardings       NA
 8 Monday 01/11/2010   952 greenAlightings      NA
 9 Monday 01/11/2010   952 greenAverage         NA
10 Monday 01/11/2010   952 bannerBoardings      NA
# ℹ 13,742 more rows
```

Reshaping data from **wide to long**

There are many ways to select the columns we want. Use `?tidyr_tidy_select` to look at more column selection options.

```
long <- circ %>%
  pivot_longer( !c(day, date, daily))
long
```

```
# A tibble: 13,752 × 5
   day    date       daily name              value
   <chr>  <chr>      <dbl> <chr>             <dbl>
 1 Monday 01/11/2010   952 orangeBoardings     877
 2 Monday 01/11/2010   952 orangeAlightings   1027
 3 Monday 01/11/2010   952 orangeAverage       952
 4 Monday 01/11/2010   952 purpleBoardings      NA
 5 Monday 01/11/2010   952 purpleAlightings     NA
 6 Monday 01/11/2010   952 purpleAverage        NA
 7 Monday 01/11/2010   952 greenBoardings       NA
 8 Monday 01/11/2010   952 greenAlightings      NA
 9 Monday 01/11/2010   952 greenAverage         NA
10 Monday 01/11/2010   952 bannerBoardings      NA
# ℹ 13,742 more rows
```

Cleaning up long data

We will use `str_replace` from the `stringr` package to put `_` in the names

```
long <- long %>% mutate(
  name = str_replace(name, "Board", "_Board"),
  name = str_replace(name, "Alight", "_Alight"),
  name = str_replace(name, "Average", "_Average")
)
long
```

```
# A tibble: 13,752 × 5
   day    date       daily name                value
   <chr>  <chr>      <dbl> <chr>               <dbl>
 1 Monday 01/11/2010   952 orange_Boardings      877
 2 Monday 01/11/2010   952 orange_Alightings    1027
 3 Monday 01/11/2010   952 orange_Average        952
 4 Monday 01/11/2010   952 purple_Boardings       NA
 5 Monday 01/11/2010   952 purple_Alightings      NA
 6 Monday 01/11/2010   952 purple_Average         NA
 7 Monday 01/11/2010   952 green_Boardings        NA
 8 Monday 01/11/2010   952 green_Alightings       NA
 9 Monday 01/11/2010   952 green_Average          NA
10 Monday 01/11/2010   952 banner_Boardings       NA
# ℹ 13,742 more rows
```

Cleaning up long data

Now each `var` is Boardings, Averages, or Alightings. We use "`into =`" to name the new columns and "`sep =`" to show where the separation should happen.

```
long <- long %>%
  separate(name, into = c("line", "type"), sep = "_")
long
```

```
# A tibble: 13,752 × 6
   day    date       daily line   type       value
   <chr>  <chr>      <dbl> <chr>  <chr>      <dbl>
 1 Monday 01/11/2010   952 orange Boardings    877
 2 Monday 01/11/2010   952 orange Alightings  1027
 3 Monday 01/11/2010   952 orange Average      952
 4 Monday 01/11/2010   952 purple Boardings     NA
 5 Monday 01/11/2010   952 purple Alightings    NA
 6 Monday 01/11/2010   952 purple Average       NA
 7 Monday 01/11/2010   952 green  Boardings     NA
 8 Monday 01/11/2010   952 green  Alightings    NA
 9 Monday 01/11/2010   952 green  Average       NA
10 Monday 01/11/2010   952 banner Boardings     NA
# ℹ 13,742 more rows
```

GUT CHECK!

What does `pivot_longer()` do?

A. Summarize data

B. Subsets data

C. Reshape data

pivot_wider...

Reshaping data from **long to wide**

`pivot_wider()` - spreads row data into columns (`tidyr` package)

- `names_from =` the old column whose contents will be spread into multiple new column names.
- `values_from =` the old column whose contents will fill in the values of those new columns.

```
{wide_data} <- {long_data} %>%
  pivot_wider(names_from = {Old column name: contains new column names},
              values_from = {Old column name: contains new cell values})
```

Reshaping data from **long to wide**

```
long_data
```

```
# A tibble: 3 × 2
  Month             Rate
  <chr>            <dbl>
1 June_vacc_rate  0.516
2 May_vacc_rate   0.514
3 April_vacc_rate 0.511
```

```
wide_data <- long_data %>% pivot_wider(names_from = "Month",
                                       values_from = "Rate")
wide_data
```

```
# A tibble: 1 × 3
  June_vacc_rate May_vacc_rate April_vacc_rate
           <dbl>         <dbl>           <dbl>
1          0.516         0.514           0.511
```

Reshaping Charm City Circulator

```
long
```

```
# A tibble: 13,752 × 6
   day    date       daily line   type        value
   <chr>  <chr>      <dbl> <chr>  <chr>       <dbl>
 1 Monday 01/11/2010   952 orange Boardings     877
 2 Monday 01/11/2010   952 orange Alightings   1027
 3 Monday 01/11/2010   952 orange Average       952
 4 Monday 01/11/2010   952 purple Boardings      NA
 5 Monday 01/11/2010   952 purple Alightings     NA
 6 Monday 01/11/2010   952 purple Average        NA
 7 Monday 01/11/2010   952 green  Boardings      NA
 8 Monday 01/11/2010   952 green  Alightings     NA
 9 Monday 01/11/2010   952 green  Average        NA
10 Monday 01/11/2010   952 banner Boardings      NA
# ℹ 13,742 more rows
```

Reshaping Charm City Circulator

```
wide <- long %>% pivot_wider(names_from = "type",
                             values_from = "value")
wide
```

```
# A tibble: 4,584 × 7
   day       date       daily line   Boardings Alightings Average
   <chr>     <chr>      <dbl> <chr>      <dbl>      <dbl>   <dbl>
 1 Monday    01/11/2010  952  orange       877       1027    952 
 2 Monday    01/11/2010  952  purple        NA         NA     NA 
 3 Monday    01/11/2010  952  green         NA         NA     NA 
 4 Monday    01/11/2010  952  banner        NA         NA     NA 
 5 Tuesday   01/12/2010  796  orange       777        815    796 
 6 Tuesday   01/12/2010  796  purple        NA         NA     NA 
 7 Tuesday   01/12/2010  796  green         NA         NA     NA 
 8 Tuesday   01/12/2010  796  banner        NA         NA     NA 
 9 Wednesday 01/13/2010 1212. orange      1203       1220   1212.
10 Wednesday 01/13/2010 1212. purple        NA         NA     NA 
# ℹ 4,574 more rows
```

Adding prefixes

Prefixes when pivoting

the `datasets::airquality` data shows various air quality metrics measured in New York in 1973.

```
air <- datasets::airquality %>% select(Temp, Month, Day)
air
```

```
    Temp Month Day
1     67     5   1
2     72     5   2
3     74     5   3
4     62     5   4
5     56     5   5
6     66     5   6
7     65     5   7
8     59     5   8
9     61     5   9
10    69     5  10
11    74     5  11
12    69     5  12
13    66     5  13
14    68     5  14
15    58     5  15
16    64     5  16
17    66     5  17
18    57     5  18
19    68     5  19
20    62     5  20
21    59     5  21
```

Prefixes when pivoting

Let’s pivot `Month` wider: but it might be helpful to add “Month” to the new column name so it isn’t just numbers.

```
air %>% pivot_wider(names_from = "Month",
                    values_from = "Temp")
```

```
# A tibble: 31 × 6
     Day   `5`   `6`   `7`   `8`   `9`
   <int> <int> <int> <int> <int> <int>
 1     1    67    78    84    81    91
 2     2    72    74    85    81    92
 3     3    74    67    81    82    93
 4     4    62    84    84    86    93
 5     5    56    85    83    85    87
 6     6    66    79    83    87    84
 7     7    65    82    88    89    80
 8     8    59    87    92    90    78
 9     9    61    90    92    90    75
10    10    69    87    89    92    73
# ℹ 21 more rows
```

Prefixes when pivoting

Much better!

```
air %>% pivot_wider(names_from = "Month",
                    values_from = "Temp",
                    names_prefix = "Month_")
```

```
# A tibble: 31 × 6
     Day Month_5 Month_6 Month_7 Month_8 Month_9
   <int>   <int>   <int>   <int>   <int>   <int>
 1     1      67      78      84      81      91
 2     2      72      74      85      81      92
 3     3      74      67      81      82      93
 4     4      62      84      84      86      93
 5     5      56      85      83      85      87
 6     6      66      79      83      87      84
 7     7      65      82      88      89      80
 8     8      59      87      92      90      78
 9     9      61      90      92      90      75
10    10      69      87      89      92      73
# ℹ 21 more rows
```

Summary

- `tidyr` package helps us convert between wide and long data
- `pivot_longer()` goes from wide -> long
  - Specify columns you want to pivot
  - Specify `names_to =` and `values_to =` for custom naming
- `pivot_wider()` goes from long -> wide
  - Specify `names_from =` and `values_from =`

https://sisbid.github.io/Data-Wrangling/labs/data-reshaping-lab.Rmd